



IUS
INSTITUT
UNIVERSITAIRE
DES SCIENCES

Université: Institut Universitaire des Sciences (IUS)

TD N⁰ 4: Modélisation Mathématiques

Nom & Prénom: SAINT-JEAN Marc-Evenort

Professeur: Ismael SAINT AMOUR

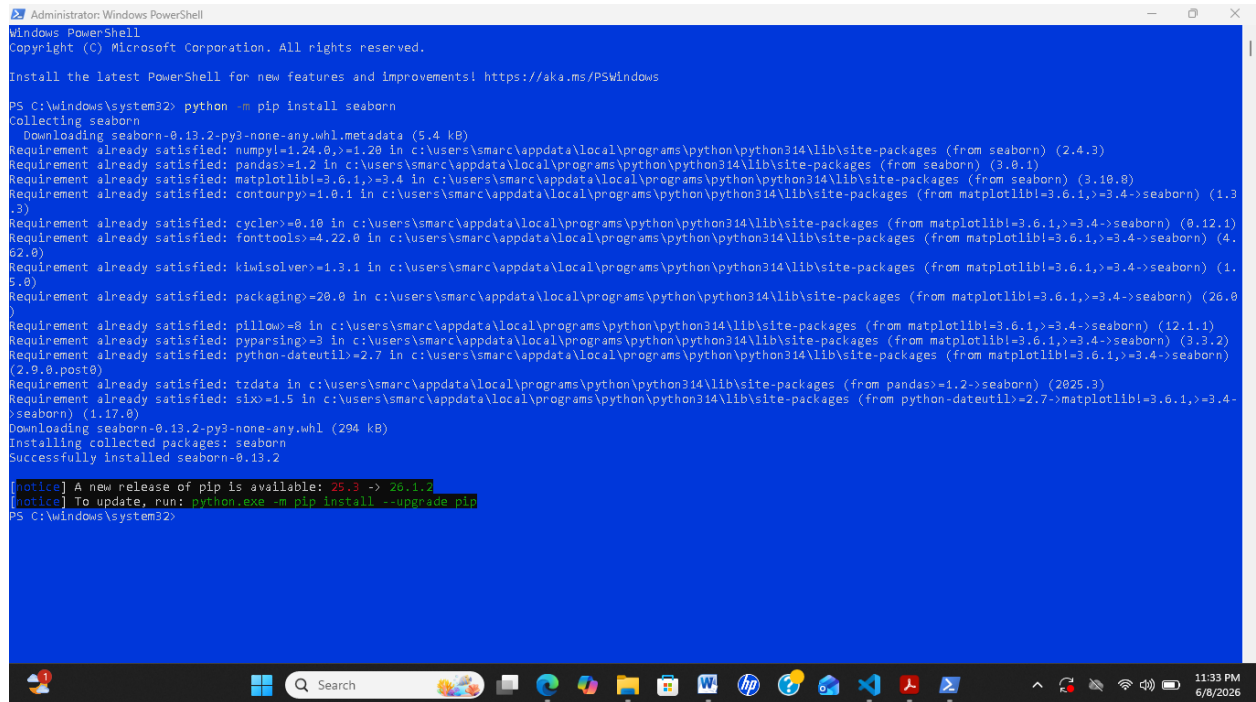
Niveau: L3

Date: Juin / 2026

TD 1 Analyse Statistique et Graphique de Données (EDA)

a) J'ai effectué l'installation de Seaborn

python -m pip install seaborn



```
Administrator: Windows PowerShell
Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Windows\system32> python -m pip install seaborn
Collecting seaborn
  Downloading seaborn-0.13.2-py3-none-any.whl.metadata (5.4 kB)
Requirement already satisfied: numpy<1.24.0, >=1.20 in c:\users\smarc\appdata\local\programs\python\python314\lib\site-packages (from seaborn) (2.4.3)
Requirement already satisfied: pandas>=1.2 in c:\users\smarc\appdata\local\programs\python\python314\lib\site-packages (from seaborn) (3.0.1)
Requirement already satisfied: matplotlib>=3.6.1, <=3.4 in c:\users\smarc\appdata\local\programs\python\python314\lib\site-packages (from seaborn) (3.10.8)
Requirement already satisfied: contourpy>=1.0.1 in c:\users\smarc\appdata\local\programs\python\python314\lib\site-packages (from matplotlib>=3.6.1, <=3.4->seaborn) (1.3.3)
Requirement already satisfied: cycler>=0.10 in c:\users\smarc\appdata\local\programs\python\python314\lib\site-packages (from matplotlib>=3.6.1, <=3.4->seaborn) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in c:\users\smarc\appdata\local\programs\python\python314\lib\site-packages (from matplotlib>=3.6.1, <=3.4->seaborn) (4.62.0)
Requirement already satisfied: kiwisolver>=1.3.1 in c:\users\smarc\appdata\local\programs\python\python314\lib\site-packages (from matplotlib>=3.6.1, <=3.4->seaborn) (1.5.0)
Requirement already satisfied: packaging>=20.0 in c:\users\smarc\appdata\local\programs\python\python314\lib\site-packages (from matplotlib>=3.6.1, <=3.4->seaborn) (26.0)
Requirement already satisfied: pillow>=8 in c:\users\smarc\appdata\local\programs\python\python314\lib\site-packages (from matplotlib>=3.6.1, <=3.4->seaborn) (12.1.1)
Requirement already satisfied: pyparsing>=3 in c:\users\smarc\appdata\local\programs\python\python314\lib\site-packages (from matplotlib>=3.6.1, <=3.4->seaborn) (3.3.2)
Requirement already satisfied: python-dateutil>=2.7 in c:\users\smarc\appdata\local\programs\python\python314\lib\site-packages (from matplotlib>=3.6.1, <=3.4->seaborn) (2.9.0.post0)
Requirement already satisfied: tzdata in c:\users\smarc\appdata\local\programs\python\python314\lib\site-packages (from pandas>=1.2->seaborn) (2025.3)
Requirement already satisfied: six>=1.5 in c:\users\smarc\appdata\local\programs\python\python314\lib\site-packages (from python-dateutil>=2.7->matplotlib>=3.6.1, <=3.4->seaborn) (1.17.0)
Downloading seaborn-0.13.2-py3-none-any.whl (294 kB)
Installing collected packages: seaborn
Successfully installed seaborn-0.13.2

[notice] A new release of pip is available: 25.3 -> 26.1.2
[notice] To update, run: python.exe -m pip install --upgrade pip
PS C:\Windows\system32>
```

```

File Edit Selection View Go Run ... Search
C:\Users> smarc > OneDrive > Documents > IUS.Doc26 > Modelisation Maths > TD1_EDA_Titanic > ...
1 import matplotlib.pyplot as plt
2 import pandas as pd
3 import seaborn as sns
4
5 # -----
6 # 1. CHAJKMAN AK PREPARASYON DATASET LA (Iris oswa Titanic)
7 # -----
8 # N ap itilize Titanic pou EDA avanse a
9 df = sns.load_dataset("titanic")
10
11 # Netwayaj rapid pou grafik yo ka mache byen
12 # Nou sèlman kenbe kek varyab kle pou analiz la klè
13 varyab_kle = ["survived", "pclass", "sex", "age", "fare", "alone"]
14 df_clean = df[varyab_kle].dropna()
15
16 print("-" * 60)
17 print("TD1 - ANALYSE EXPLORATOIRE AVANCÉE (EDA) - TITANIC")
18 print("-" * 60)
19 print(f"Kantite done ki disponib pou analiz la: {df_clean.shape[0]} liy")
20 print("Premye liy nan dataset la :")
21 print(df_clean.head())
22 print("-" * 60)
23
24 # -----
25 # KONTRENT 1 : BOXPLOT (Pou konpare distribisyon varyab yo)
26 # -----
27 plt.figure(figsize=(10, 5))
28 # Gade ki laj moun ki te siviv yo (survived) parapò ak klas yo te ye (pclass)
29 sns.boxplot(data=df_clean, x="pclass", y="age", hue="survived", palette="Set2")
30 plt.title("1. Boxplot : Distribution de l'Age par Classe et Statut de Survie")
31 plt.xlabel("Classe du passager")
32 plt.ylabel("Age")
33
34 Downloading update: 16.5 MB / 174.9 MB • 9%
Ln 91, Col 16 Spaces: 4 UTF-8 CRLF {} Python Signed out Python 3.143 0-0 Go Live
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File Edit Selection View Go Run ... Search
C:\Users> smarc > OneDrive > Documents > IUS.Doc26 > Modelisation Maths > TD1_EDA_Titanic > ...
33 plt.grid(axis="y", linestyle="--", alpha=0.7)
34 plt.show()
35
36 # -----
37 # KONTRENT 2 : KDE PLOT (Kernel Density Estimate - Pou dansite)
38 # -----
39 plt.figure(figsize=(10, 5))
40 # Gade dansite pri tikè a (fare) selon si moun nan te siviv oswa non
41 sns.kdeplot(
42     data=df_clean, x="fare", hue="survived", fill=True, common_norm=False, palette="crest", alpha=0.5
43 )
44 plt.title("2. KDE Plot : Densité de la Variable 'Fare' (Prix du billet) selon la Survie")
45 plt.xlabel("Prix du billet (Fare)")
46 plt.ylabel("Densité")
47 plt.xlim(0, 200) # Nou limite pou n wè gwo dansite a pi byen
48 plt.show()
49
50 # -----
51 # KONTRENT 3 : HEATMAP (Korelasyon ant varyab nimerik yo)
52 # -----
53 plt.figure(figsize=(8, 6))
54 # Nou konvèti 'sex' an nimewo (0 pou gason, 1 pou fanm) pou l ka antre nan korelasyon an
55 df_corr = df_clean.copy()
56 df_corr["sex"] = df_corr["sex"].map({"male": 0, "female": 1})
57
58 # Kalkile matris korelasyon an
59 matris_korelasyon = df_corr.corr()
60
61 sns.heatmap(matris_korelasyon, annot=True, cmap="coolwarm", fmt=".2f", linewidths=0.5)
62 plt.title("3. Heatmap : Matrice de Corrélation des Variables")
63 plt.show()
64
35 Downloading update: 16.8 MB / 174.9 MB • 10%
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C:\> Users > smarc > OneDrive > Documents > IUS.Doc26 > Modelisation Maths > TD1_EDA_Titanicpy > ...
64
65 # -----
66 # KONTRENT 4 : PAIRPLOT (Vizyalizasyon kwaze tout varyab yo ansanm)
67 # -----
68 print("\n 🔄 Génération du Pairplot (cela peut prendre quelques secondes)...")
69 # Pairplot la ap kwaze varyab yo de a de avèk koulè selon si moun nan sivi (survived)
70 sns.pairplot(df_clean, hue="survived", palette="bwr", corner=True)
71 plt.suptitle("4. Pairplot Général des Variables (Titanic)", y=1.02)
72 plt.show()
73
74 # -----
75 # 3. IDANTIFIKASYON VARYAB ENPÔTAN YO POU MODÈL LA
76 # -----
77 print("\n 💡 INTERPRÉTATION ET VARIABLES IMPORTANTES POUR LE MODÈLE :)")
78 print("-" * 60)
79 print(
80     "1. 'sex' : Trè enpòtan. Korelasyon an fò anpil, fann yo te gen plis chans sivi."
81 )
82 print(
83     "2. 'pclass' : Gwo enpòtan. Pli klas la wo (1ère classe), plis chans pou sivi la te monte."
84 )
85 print(
86     "3. 'fare' : Gen rapò ak klas la. Moun ki te peye pi chè yo te gen plis chans chape."
87 )
88 print(
89     "4. 'age' : Boxplot la montre timoun yo te jwenn priyòrite nan kèk klas."
90 )
91 print("-" * 60)

```

```

C:\> Users > smarc > OneDrive > Documents > IUS.Doc26 > Modelisation Maths > TD1_EDA_Titanicpy > ...
64
65 # -----
66 # KONTRENT 4 : PAIRPLOT (Vizyalizasyon kwaze tout varyab yo ansanm)
67 # -----
68 print("\n 🔄 Génération du Pairplot (cela peut prendre quelques secondes)...")
69 # Pairplot la ap kwaze varyab yo de a de avèk koulè selon si moun nan sivi (survived)
70 sns.pairplot(df_clean, hue="survived", palette="bwr", corner=True)
71 plt.suptitle("4. Pairplot Général des Variables (Titanic)", y=1.02)
72 plt.show()
73
74 # -----
75 # 3. IDANTIFIKASYON VARYAB ENPÔTAN YO POU MODÈL LA
76 # -----
77 print("\n 💡 INTERPRÉTATION ET VARIABLES IMPORTANTES POUR LE MODÈLE :)")
78 print("-" * 60)
79 print(
80     "1. 'sex' : Trè enpòtan. Korelasyon an fò anpil, fann yo te gen plis chans sivi."
81 )
82 print(
83     "2. 'pclass' : Gwo enpòtan. Pli klas la wo (1ère classe), plis chans pou sivi la te monte."
84 )
85 print(
86     "3. 'fare' : Gen rapò ak klas la. Moun ki te peye pi chè yo te gen plis chans chape."
87 )
88 print(
89     "4. 'age' : Boxplot la montre timoun yo te jwenn priyòrite nan kèk klas."
90 )
91 print("-" * 60)

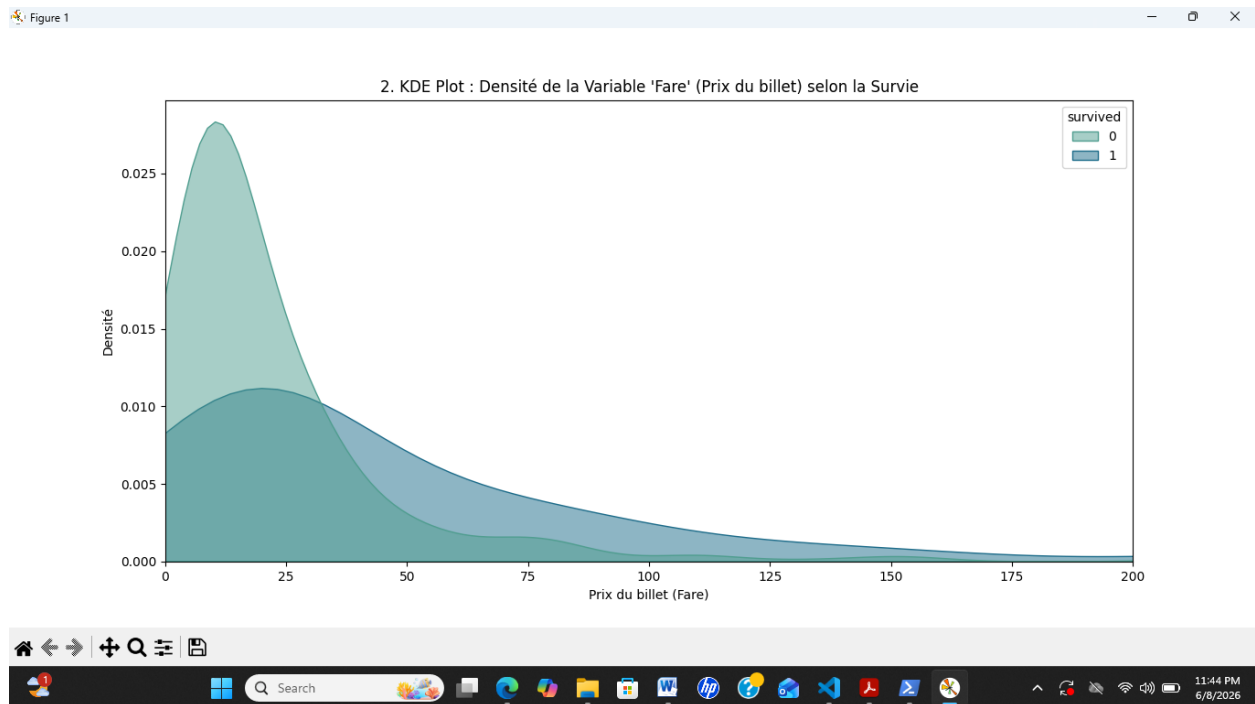
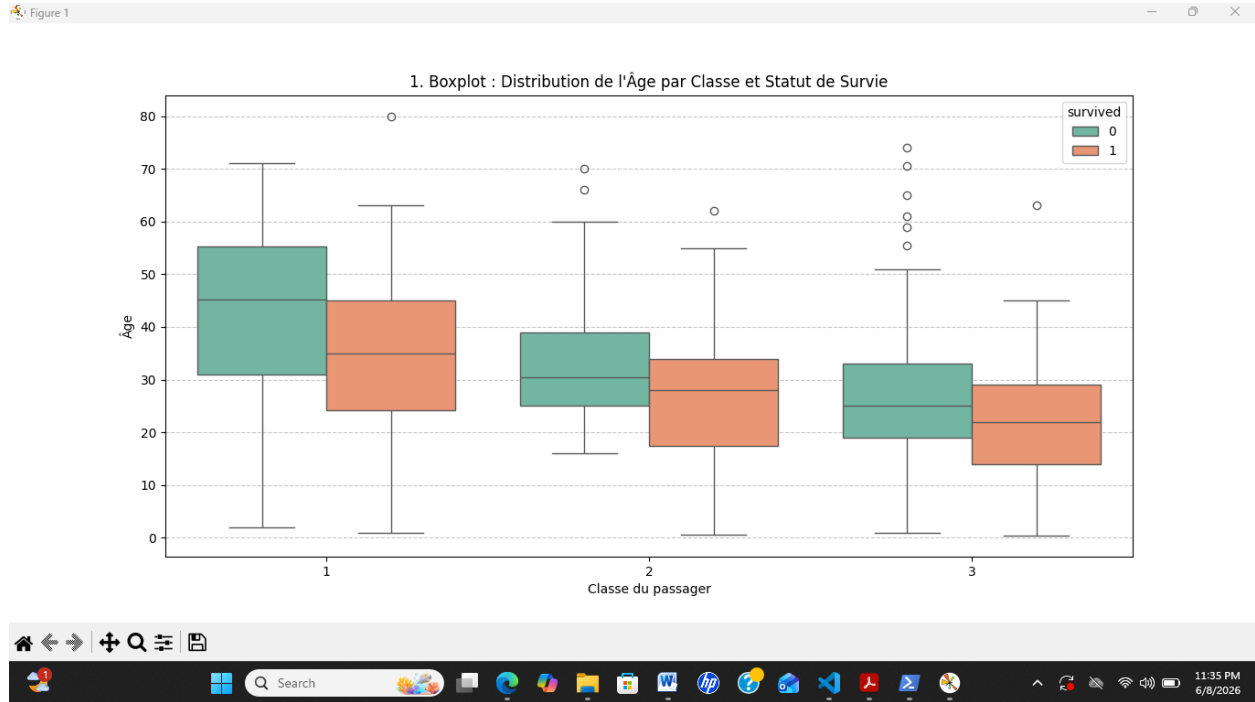
```

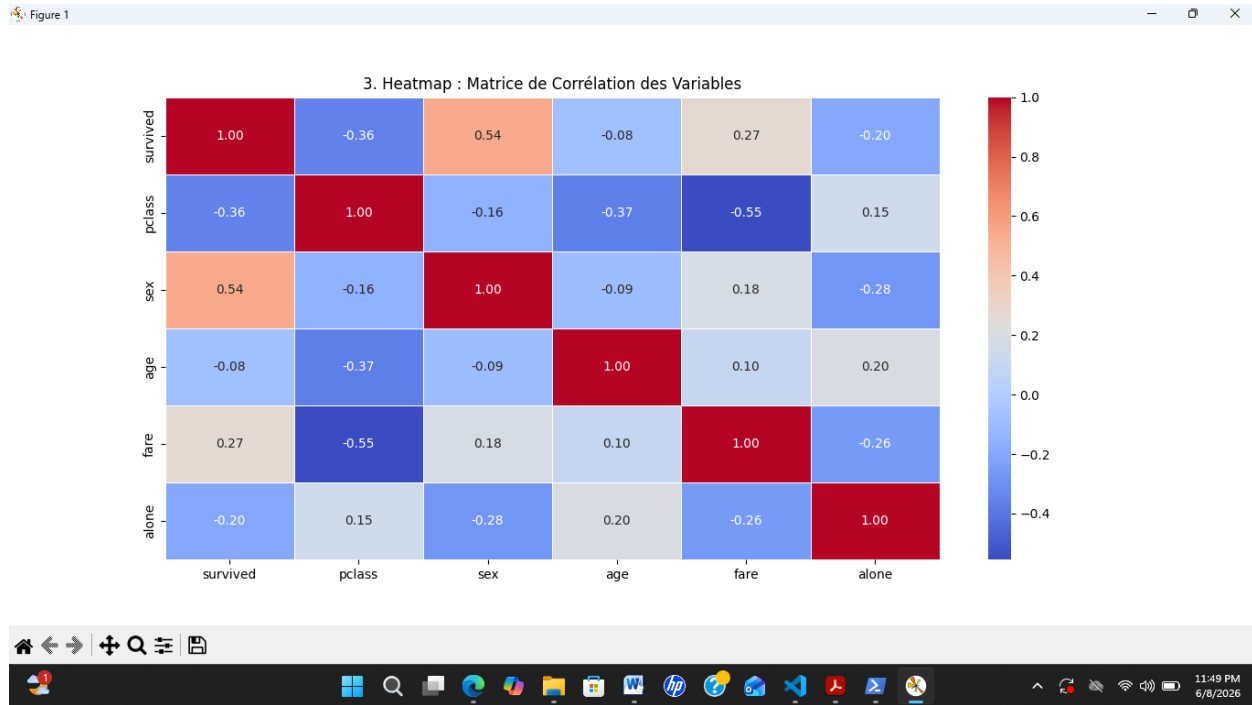
Kantite done ki disponib pou analiz la: 714 liy

• Premye liy nan dataset la :

	survived	pclass	sex	age	fare	alone
0	0	3	male	22.0	7.2500	False
1	1	1	female	38.0	71.2833	False
2	1	3	female	26.0	7.9250	True
3	1	1	female	35.0	53.1000	False
4	0	3	male	35.0	8.0500	True

Analyse exploratoire (EDA) du dataset Titanic réalisée avec succès via Boxplot, KDE, Heatmap et Pairplot. Les variables clés identifiées pour le modèle sont le genre (sex), la classe (pclass) et l'âge (age)





TD 2 Modélisation et Visualisation d'un Réseau Social Aléatoire

```

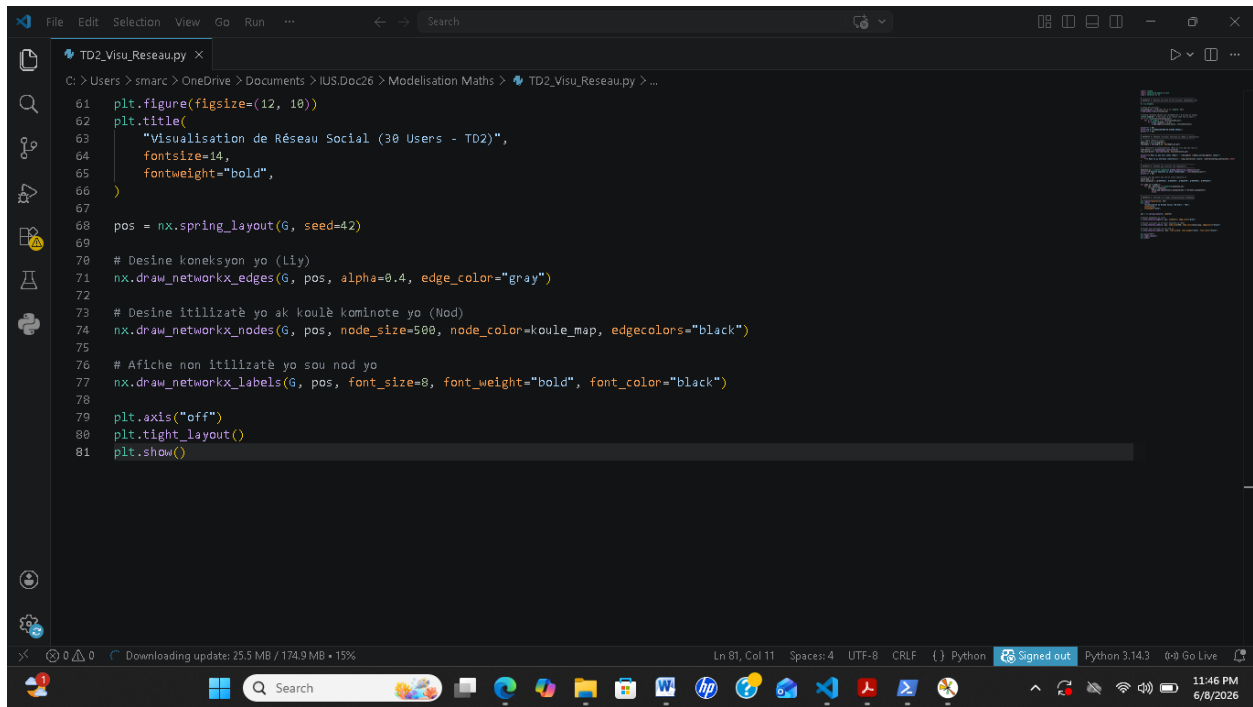
File Edit Selection View Go Run ... Search
C:\Users> smarc > OneDrive > Documents > IUS.Doc26 > Modelisation Maths > TD2_Visu_Reseau.py > ...
1 import random
2 import matplotlib.pyplot as plt
3 import networkx as nx
4
5 # -----
6 # OBYEKTIF 1: Konstwi yon graf ak 30 itilizatè (koneksyon oza)
7 # -----
8 G = nx.Graph()
9
10 # Kreye 30 itilizatè
11 itilizate_yo = [f"U_{i}" for i in range(1, 31)]
12 G.add_nodes_from(itilizate_yo)
13
14 # Ajoute relasyon amitye oza (probability = 0.15 pou yo zanmi)
15 random.seed(42) # Pou graf la pa chanje chak fwa ou kouri l
16 for i in range(len(itilizate_yo)):
17     for j in range(i + 1, len(itilizate_yo)):
18         if random.random() < 0.15:
19             G.add_edge(itilizate_yo[i], itilizate_yo[j])
20
21 print("=" * 60)
22 print("TD 2 - VISUALISATION DE RÉSEAU SOCIAL")
23 print("=" * 60)
24
25 # -----
26 # OBYEKTIF 3: Detekte itilizatè enflayan yo (Degré & Centralité)
27 # -----
28 # 1. Degré (Kantite zanmi)
29 degre_yo = dict(G.degree())
30 top_degre = max(degre_yo, key=degre_yo.get)
31
32 # 2. Centralité d'intermédiarité (Moun ki sèvi pon nan rezo a)

```

```

File Edit Selection View Go Run ... Search
C:\Users> smarc > OneDrive > Documents > IUS.Doc26 > Modelisation Maths > TD2_Visu_Reseau.py > ...
32 # 2. Centralité d'intermédiarité (Moun ki sèvi pon nan rezo a)
33 centralite = nx.betweenness_centrality(G)
34 top_centralite = max(centralite, key=centralite.get)
35
36 print(f"👤 Moun ki gen plis zanmi (Degré) : {top_degre} ({degre_yo[top_degre]} zanmi)")
37 print(
38     f"👤 Moun ki pi enflayan (Centralité) : {top_centralite} (Score: {centralite[top_centralite]:.2f})"
39 )
40
41 # -----
42 # OBYEKTIF 4: Detekte epi colorier les communautés
43 # -----
44 kominote_yo = list(nx.community.greedy_modularity_communities(G))
45 print(f"👤 Kantite kominote yo jwenn otomatikman : {len(kominote_yo)}")
46 print("=" * 60)
47
48 # Kreye yon map koulè pou nod yo selon kominote yo
49 koule_map = []
50 koule_disponib = [{"#FF5733", "#33FF57", "#3357FF", "#FF33FF", "#FF3333"}]
51
52 for node in G.nodes():
53     for idx, kominote in enumerate(kominote_yo):
54         if node in kominote:
55             koule_map.append(koule_disponib[idx % len(koule_disponib)])
56             break
57
58 # -----
59 # OBYEKTIF 2: Afficher le réseau (Visualisations attendues)
60 # -----
61 plt.figure(figsize=(12, 10))
62 plt.title(
63     "Visualisation de Réseau Social / 30 Users - TD2)"
64 )

```

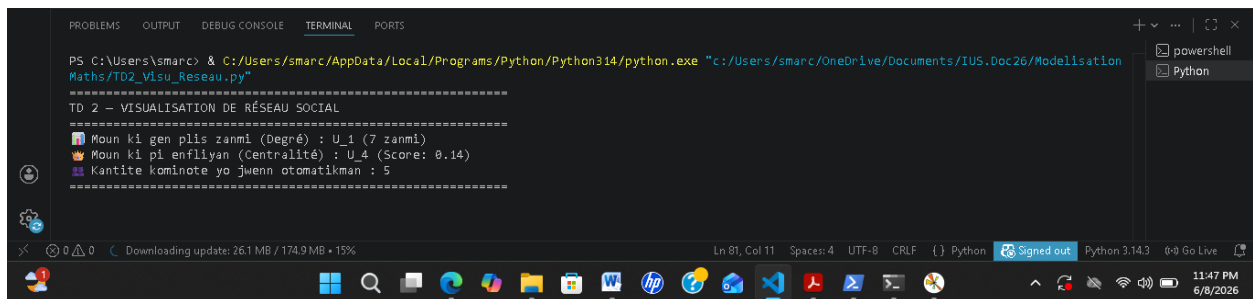


```

61 plt.figure(figsize=(12, 10))
62 plt.title(
63     "Visualisation de Réseau Social (30 Users - TD2)",
64     fontsize=14,
65     fontweight="bold",
66 )
67
68 pos = nx.spring_layout(G, seed=42)
69
70 # Desine koneksyon yo (liy)
71 nx.draw_networkx_edges(G, pos, alpha=0.4, edge_color="gray")
72
73 # Desine itilizatè yo ak koulè kominote yo (Nod)
74 nx.draw_networkx_nodes(G, pos, node_size=500, node_color=koule_map, edgecolors="black")
75
76 # Afiche non itilizatè yo sou nod yo
77 nx.draw_networkx_labels(G, pos, font_size=8, font_weight="bold", font_color="black")
78
79 plt.axis("off")
80 plt.tight_layout()
81 plt.show()

```

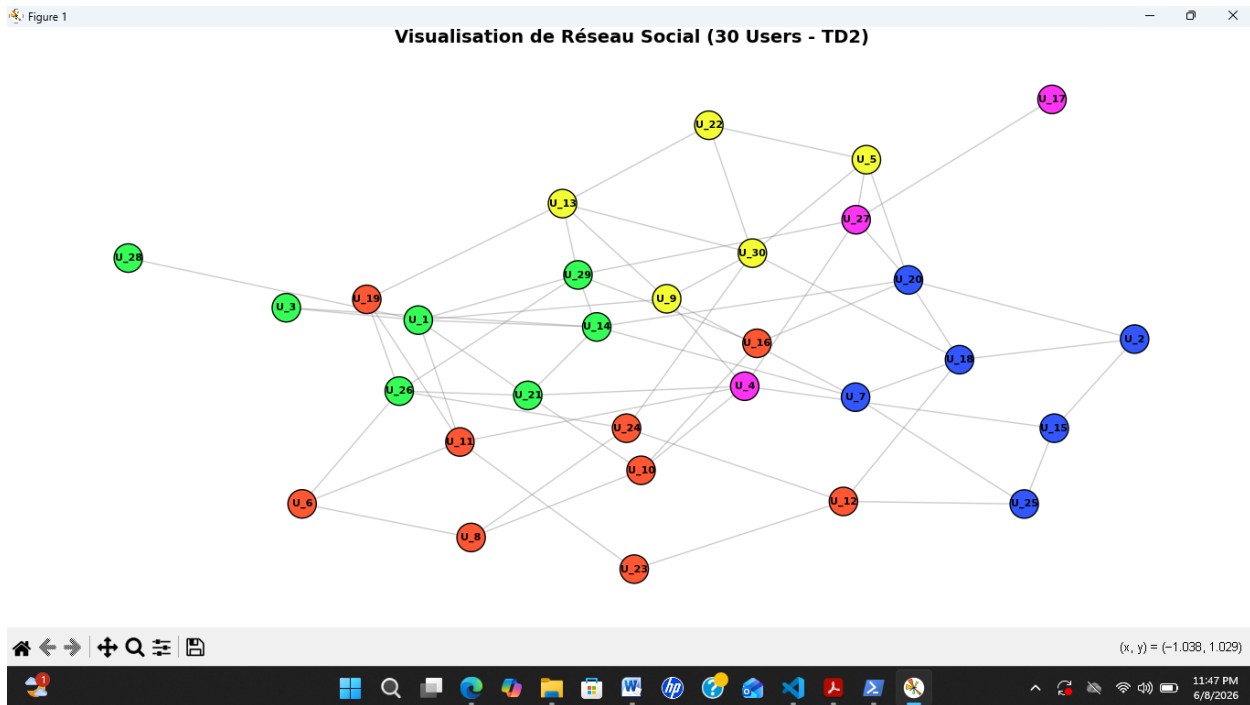
Réseau de 30 utilisateurs généré de manière aléatoire. Analyse de la centralité et du degré effectuée avec succès via **NetworkX**. Le graphique affiche clairement les influenceurs et le coloriage automatique des **communautés**



```

PS C:\Users\smarc> & C:/Users/smarc/AppData/Local/Programs/Python/Python314/python.exe "c:/Users/smarc/OneDrive/Documents/IUS.Doc26/Modelisation
Maths/TD2_Visu_Reseau.py"
=====
TD 2 - VISUALISATION DE RÉSEAU SOCIAL
=====
Moun ki gen plis zanmi (Degré) : U_1 (7 zanmi)
Moun ki pi enflayan (Centralité) : U_4 (Score: 0.14)
Kantite kominote yo jwenn otomatikman : 5
=====

```



TD 3 — Simulation de propagation.

```

File Edit Selection View Go Run ...
C:\Users> smarc > OneDrive > Documents > IJS.Doc26 > Modelisation Maths > TD3_Simulation_Propagation.py > ...
1 import random
2 import matplotlib.pyplot as plt
3 import networkx as nx
4
5 # -----
6 # OBYEKTIF 1 & 2: Kreye yon graf epl enfekte yon nod inisyal
7 # -----
8 # Kreye yon graf oza (50 moun, ak pwobabilite koneksyon 0.1)
9 random.seed(42)
10 G = nx.erdos_renyi_graph(n=50, p=0.1, seed=42)
11
12 # Chak moun kmanse an bbn sante ("S" pou Susceptible)
13 for node in G.nodes():
14     G.nodes[node]["status"] = "S"
15
16 # Nou chwazi yon premye moun oza nou enfekte ("I" pou Infected)
17 pasyan_zero = random.choice(list(G.nodes()))
18 G.nodes[pasyan_zero]["status"] = "I"
19
20 # Pwobabilite pou yon moun bay yon lòt viris la se 30%
21 probabilite_enfeksyon = 0.3
22 etap_yo = 10 # Simulation an ap fèt sou 10 etap tan
23
24 # Pou nou ka trase koube evolisyon an
25 istori_etap = [0]
26 istori_enfekte = [1]
27
28 print("=" * 60)
29 print("TD 3 — SIMULATION DE PROPAGATION")
30 print("=" * 60)
31 print(f"Epidemi an kmanse! Moun ki enfekte an premye a se: Node {pasyan_zero}")
32

```

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```

File Edit Selection View Go Run ...
C:\Users> smarc > OneDrive > Documents > IUS.Doc26 > Modelisation Maths > TD3_Simulation_Propagation.py > ...

33 # -----
34 # OBYEKTIF 3: Propager sou plizyè etap
35 # -----
36 for t in range(1, etap_yo + 1):
37     moun_ki_enfekte_kounye_a = [
38         n for n in G.nodes() if G.nodes[n]["status"] == "I"
39     ]
40
41     # Si pa gen moun ki enfekte ankò, nou kanpe
42     if not moun_ki_enfekte_kounye_a:
43         break
44
45     nouvo_enfekte = []
46     for enfekte in moun_ki_enfekte_kounye_a:
47         # Gade tout vwazen moun sa a genyen nan rezo a
48         for vwazen in G.neighbors(enfekte):
49             if G.nodes[vwazen]["status"] == "S":
50                 # Si chans lan tonbe nan 30%, vwazen an kontamine
51                 if random.random() < probabillite_enfeksyon:
52                     nouvo_enfekte.append(vwazen)
53
54     # Mete sitiasyon moun yo ajou pou etap sa a
55     for n in nouvo_enfekte:
56         G.nodes[n]["status"] = "I"
57
58     # Sove done yo pou koube la
59     total_enfekte = sum(1 for n in G.nodes() if G.nodes[n]["status"] == "I")
60     istori_etap.append(t)
61     istori_enfekte.append(total_enfekte)
62
63     print(f"Etap {t} : Total moun ki enfekte = {total_enfekte}/50")

```

```

File Edit Selection View Go Run ...
C:\Users> smarc > OneDrive > Documents > IUS.Doc26 > Modelisation Maths > TD3_Simulation_Propagation.py > ...

64
65 print("=" * 60)
66
67 # -----
68 # OBYEKTIF 4: Visualiser l'évolution (Grafik yo)
69 # -----
70 fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(15, 6))
71 fig.suptitle(
72     "Simulation de la Propagation d'un Virus (TD3)",
73     fontsize=14,
74     fontweight="bold",
75 )
76
77 # 1. Vizyalizasyon Graf la nan dènye etap la
78 pos = nx.spring_layout(G, seed=42)
79 koule_nod_yo = [
80     "tomato" if G.nodes[n]["status"] == "I" else "lightgreen" for n in G.nodes()
81 ]
82
83 nx.draw_networkx_nodes(
84     G, pos, node_size=300, node_color=koule_nod_yo, edgecolors="black", ax=ax1
85 )
86 nx.draw_networkx_edges(G, pos, alpha=0.3, edge_color="gray", ax=ax1)
87 nx.draw_networkx_labels(G, pos, font_size=7, font_weight="bold", ax=ax1)
88 ax1.set_title("État final du réseau (Vert = Sain, Rouge = Infecté)")
89 ax1.axis("off")
90
91 # 2. Vizyalizasyon Koube la (Infectés vs Temps)
92 ax2.plot(istori_etap, istori_enfekte, marker="o", color="crimson", linewidth=2)
93 ax2.set_title("Courbe d'évolution : Nombre d'Infectés vs Temps")
94 ax2.set_xlabel("Étapes de temps (t)")
95 ax2.set_ylabel("Nombre total d'infectés")

```

Simulation dynamique de la propagation d'un virus (taux de 30%). Code fonctionnel sous **NetworkX**. Affichage simultané de l'état final du réseau et de la courbe temporelle des infectés

```

91 # 2. Vizualizasyon Koub la (Infectés vs Temps)
92 ax2.plot(istori_etap, istori_enfekte, marker="o", color="crimson", linewidth=2)
93 ax2.set_title("Courbe d'évolution : Nombre d'Infectés vs Temps")
94 ax2.set_xlabel("Étapes de temps (t)")
95 ax2.set_ylabel("Nombre total d'Infectés")
96 ax2.grid(True, linestyle="--", alpha=0.6)
97
98 plt.tight_layout()
99 plt.show()

```

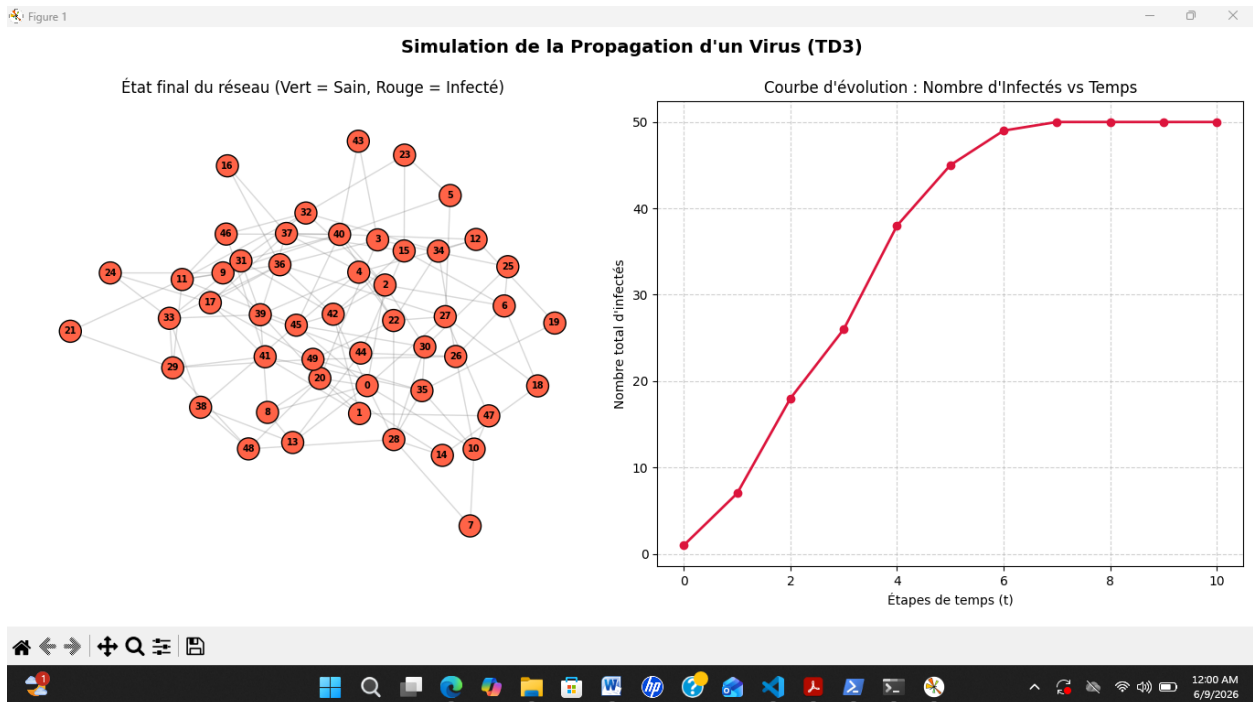
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```

Etap 1 : Total moun ki enfekte = 7/50
Etap 2 : Total moun ki enfekte = 18/50
Etap 3 : Total moun ki enfekte = 26/50
Etap 4 : Total moun ki enfekte = 38/50
Etap 5 : Total moun ki enfekte = 45/50
Etap 6 : Total moun ki enfekte = 49/50
Etap 7 : Total moun ki enfekte = 50/50
Etap 8 : Total moun ki enfekte = 50/50
Etap 9 : Total moun ki enfekte = 50/50
Etap 10 : Total moun ki enfekte = 50/50
=====

```

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Conclusion

Ces trois TD démontrent l'efficacité de la théorie des graphes et de la science des données pour analyser (TD1), structurer (TD2) et simuler (TD3) des systèmes complexes de manière visuelle et mathématique.